

# VEDHAVARSHINI N

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## Professional Summary

Electronics and Communication Engineering undergraduate with strong foundation in Embedded Systems, IoT, and PCB Design. Experienced in ESP32, Arduino, and microcontroller-based hardware development with hands-on exposure to PCB fabrication, SMT soldering, and embedded application development. Actively involved in research in DSSC and bioactive materials with publication experience. Seeking opportunities in Embedded Systems, Firmware Development, IoT, and Core Electronics domains.

## Technical Skills

- Programming : Verilog, C, Python.
- Embedded Systems : 8051 Microcontroller, Arduino, ESP32.
- Communication Protocols : UART, SPI, I2C, CAN.
- Tools & Platforms : Arduino IDE, EasyEDA, KiCad, Altium Designer, MATLAB, Quartus, Xilinx Vivado.

## Education

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|-------------------------------------------------------------------------|-------------|
| B.E Electronics and Communication Engineering                           | 2023 - 2027 |
| Easwari Engineering College, Chennai CGPA: 9.0/10.0                     |             |
| Higher Secondary Certificate (HSC)                                      | 2021 - 2023 |
| Government Girls Higher Secondary School, Chidambaram Percentage: 88.6% |             |

## Academic Projects

|                                                                                                                                                                                                                                                            |         |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| <b>Fruit Disease Detection using Deep Learning (MATLAB)</b>                                                                                                                                                                                                | 03/2026 |
| <ul style="list-style-type: none"><li>• Developed a ResNet50-based deep learning model for fruit disease classification using MATLAB with data Augmentation and transfer learning.</li></ul> Technologies: MATLAB, Deep Learning, Image Processing.        |         |
| <b>Phase-Neutral Monitoring Smart Energy Meter</b>                                                                                                                                                                                                         | 10/2025 |
| <ul style="list-style-type: none"><li>• Developed an ESP32-based IoT system to detect electricity theft via phase-neutral current comparison with real-time alert and load control.</li></ul> Technologies: ESP32, IoT Cloud, Relay, Buzzer, Arduino IDE.  |         |
| <b>Anti-Sleep Alarm Detector for Drivers</b>                                                                                                                                                                                                               | 02/2024 |
| <ul style="list-style-type: none"><li>• Developed an Arduino-based driver drowsiness detection system using an eye blink sensor with real-time alert triggering.</li></ul> Technologies: Arduino, Eye Blink Sensor, Embedded Systems, Buzzer, Arduino IDE. |         |

## Internship Experience

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| <b>Hardware &amp; Embedded Systems Intern – Inceptarc Technology</b>                                                                                                                                                                                                     | 02/2026 – Present |
| <ul style="list-style-type: none"><li>• Designed PCB layouts from schematic to Gerber generation.</li><li>• Performed SMT soldering and hardware debugging for embedded boards.</li><li>• Assisted in embedded hardware integration and prototype development.</li></ul> |                   |
| <b>Telecom Intern – NLC India Limited</b>                                                                                                                                                                                                                                | 06/2025 - 07/2025 |
| <ul style="list-style-type: none"><li>• Studied TANDEM, OFC, and VoIP communication systems.</li><li>• Observed telecom network operations and maintenance procedures.</li></ul>                                                                                         |                   |
| <b>Embedded Systems Intern – Space Zee Technologies</b>                                                                                                                                                                                                                  | 07/2025           |
| <ul style="list-style-type: none"><li>• Developed embedded applications using 8051, Arduino, and ESP32.</li><li>• Worked on sensor interfacing and microcontroller programming.</li></ul>                                                                                |                   |

## Research Experience

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| <b>Erbium Co-doped Dye-Sensitized Solar Cells (DSSC)</b>                                                                                                                                             |
| <ul style="list-style-type: none"><li>• Working on efficiency enhancement using erbium co-doping techniques.</li><li>• Hands-on experience in material preparation, coating, and analysis.</li></ul> |

## **Lithium Fluorophosphate Bioactive Glass**

- Working on development of bioactive glass materials for biomedical applications.
- Studying material characterization techniques and their relevance to sensor-based electronic applications.

## **Publications**

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- “ZrO<sub>2</sub> Functionalization on Fluorophosphate Glasses” — Under Review in *Ceramics International* (Q1 Journal).
- “DSSC Performance Enhancement via Ag–Ti–Er Co-doping” — Manuscript in Progress.

## **Achievements**

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- Received Best Paper Award at International Conference on Ultrasonics and Materials Science for Advanced Technology (ICUMSAT–2025) for research on Dye-Sensitized Solar Cells.

## **Certifications**

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- Introduction to Internet of Things – NPTEL (Elite + Silver) (2026).
- Altium Designer Essentials (03/2025).
- Python 101 for Data Science — IBM Developer Skills Network (Jul 2025).

## **Additional Information**

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- Member, Youth Red Cross (YRC), Easwari Engineering College.
- Shortlisted for Round 2 – Smartcity Challenge 2025.